draws water from the reservoir by endosmosis.

ii. Due to the continuous entry of water into chamber "A", turgor pressure is increased.

iii. Increase in turgor pressure in chamber "A" force, the mass flow of sugar solution to chamber "B" through the tube "T" along turgor pressure gradient.

iv. The movement of solute will continue till the solution in both the chambers attains the state of isotonic condition and the system becomes inactive.

v. However, if new sugar solution is added in chamber "A", the system will start to run again.

A similar analogous system as given in the experiment exists in plants:

Chamber "A" is analogous to mesophyll cells of the leaves which contain a higher concentration of food material in soluble form. In short "A" is the production point called "**source**".

Chamber "B" is analogous to cells of stem and roots where the food material is utilized. In short "B" is consumption end called "**sink**".

Tube "T" is analogous to the sieve tube of phloem.

Mesophyll cells draw water from the xylem (reservoir of the experiment) of the leaf by endosmosis leading to increase in the turgor pressure of mesophyll cell. The turgor pressure in the cells of stem and the roots are comparatively low and hence, the soluble organic solutes begin to flow *en masse* from mesophyll through the phloem to the cells of stem and roots along the gradient turgor pressure.

In the cells of stem and roots, the organic solutes are either consumed or converted into insoluble form and the excess water is released into xylem (by turgor pressure gradient) through cambium.

Merits:

i. When a woody or herbaceous plant is girdled, the sap contains high sugar containing exudates from cut end.

ii. Positive concentration gradient disappears when plants are defoliated.

Objections:

i. This hypothesis explains the unidirectional movement of solute only. However, bidirectional movement of solute is commonly observed in plants.

ii. Osmotic pressure of mesophyll cells and that of root hair do not confirm the requirements.

iii. This theory gives passive role to sieve tube and protoplasm, while some workers demonstrated the involvement of ATP.

11.8 Mineral Absorption

Minerals in soil exist in two forms, either dissolved in soil solution or adsorbed by colloidal clay particle. Previously, it was mistakenly assumed that absorption of mineral salts from soil took place along with absorption of water. But absorption of minerals and ascent of sap are identified as two independent processes. Minerals are absorbed not only by root hairs but also by the cells of epiblema.

Plasma membrane of root cells are not permeable to all ions and also all ions of same salt are not absorbed in equal rate.

Penetration and accumulation of ions into living cells or tissues from surrounding medium by crossing membrane is called **mineral absorption**. Movement of ions into and out of cells or tissues is termed as transport or **flux**. Entry of the ion into cell is called **influx** and exit is called **efflux**. Various theories have been put forward to explain this mechanism. They are categorized under passive mechanisms (without the involvement of metabolic energy) and active mechanisms (involvement of metabolic energy).

11.8.1 Passive Absorption

1. Ion-Exchange:

Ions of external soil solution were exchanged with same charged (anion for anion or cation for cation) ions of the root cells. There are two theories explaining this process of ion exchange namely:

XIth_Botany_Chapter_11_EM.indd 254 5/17/2020 9:19:02 PM